SECTION 2 · SUBSECTION 2.1 (CONT.)
COMPUTING EQUIPMENT AND SOFTWARE

2.1.4 INTRODUCTION.

The operator display is a single-page React application built by Vite and served as static files. It has one source file of consequence, *App.tsx*, holding the landing, progress, results and error screens plus an inline design system. The point to retain: navigation is local component state, not a router — *react-router-dom* is installed but never imported by the entry point (figure 2.1-3).

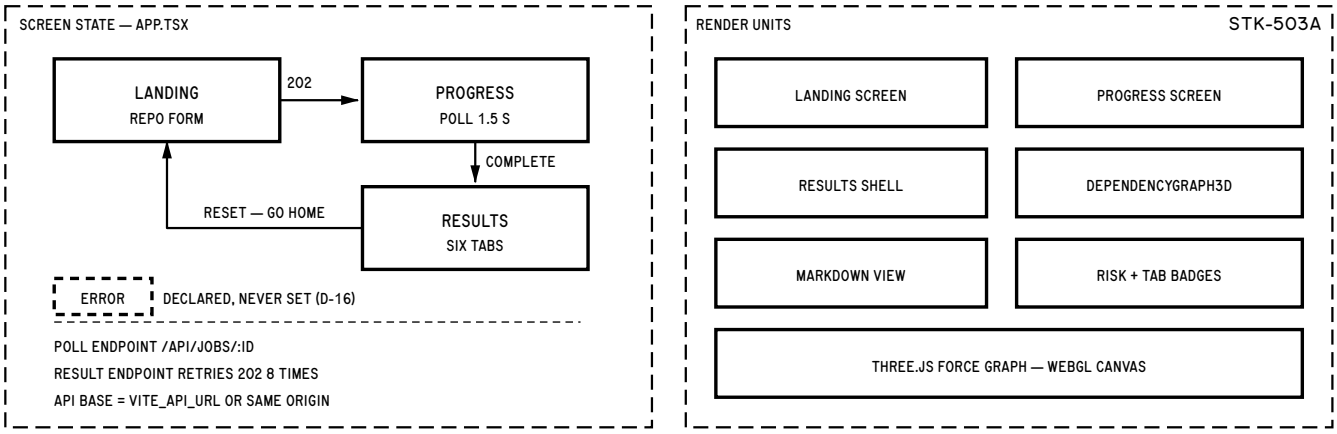


Figure 2.1-3. Screen State and Rendering Units

2.1.5 FUNCTIONAL DESCRIPTION.

main.tsx mounts the application inside a Sentry error boundary and a PostHog provider, then appends Vercel Analytics and Speed Insights. *App.tsx* submits *{repo_url, max_explanations, skip_llm}* to */api/analyze*, stores the returned job identifier, and polls */api/jobs/:id* every 1,500 milliseconds. When status becomes complete, it fetches */api/jobs/:id/result* and renders six tabs: onboarding, agent context, explanations, dependency graph, complexity and raw JSON. The graph tab hands the full node and link arrays to a WebGL force layout. A reset returns the operator to the landing form; the error screen is declared and rendered but never set, so failures surface in the progress screen (defect D-16).

2.1.5.1 Build and Headers.

Vite builds static assets into *dist/*. Vercel serves them with a Content-Security-Policy and security headers from *frontend/vercel.json*, and proxies PostHog under */ingest* and */decide*. The same file rewrites unknown paths to *index.html* — conventional for a single-page application, unnecessary while the router is unused.

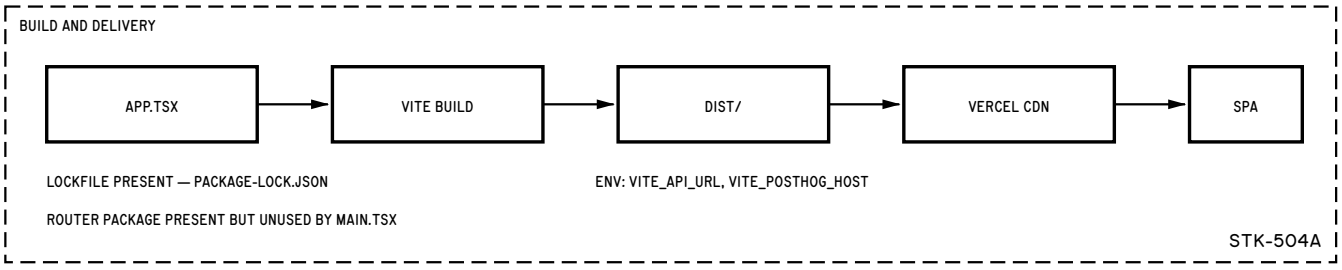


Figure 2.1-4. Build Chain and Delivery Surface

2.1.6 PERFORMANCE AND DESIGN DATA.

Table 2.1-2. Frontend Software, Declared and Resolved

Package	Declared	Resolved	Role
react / react-dom	^19.0.0	19.2.8	UI runtime
vite	^8.0.0	8.2.0	Build and dev server
typescript	^5.7.0	5.9.3	Type checking
tailwindcss	^4.0.0	4.3.3	Utility CSS
3d-force-graph	^1.80.1	1.80.1	Graph layout and camera
three	^0.179.1	0.179.1	WebGL renderer
motion	^12.0.0	12.43.0	Screen transitions
react-markdown	^10.1.0	10.1.0	Document rendering
react-router-dom	^7.18.2	7.18.2	Installed, not imported
@sentry/react	^10.73.0	10.73.0	Error reporting

CAUTION

main.tsx contains fallback Sentry and PostHog identifiers. They are not reproduced here. Treat them as public client configuration, rotate them through environment variables, and do not add server secrets to any *VITE_* variable; Vite inlines those into the bundle.

DISPLAY CHECK **OPERATOR**

```
1 SET - VITE_API_URL
2 RUN - npm run dev
3 OPEN - localhost:5173
  √ SUBMIT - job id returns
4 WATCH - progress advances
5 VERIFY - six tabs render
```

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